DigiCow Adoption Prediction Challenge – 9th Place Solution

Author: Agbaje Ayomipo
Team Name: CN-Boost

Overview

This project predicts the probability that a farmer will adopt a practice within 7 days of their first training session. The objective is to enable early identification of farmers likely to turn training into action, allowing DigiCow to prioritise

Final Position: 9th Place
Total Pipeline Runtime: ~1 hour 30 minutes

Libraries Used

numpy==2.2.6
pandas==2.3.0
tqdm==4.67.1
scikit-learn==1.7.0
feature-engine==1.8.3
xgboost==3.0.2
catboost==1.2.8
lightgbm==4.6.0
scipy==1.16.0
sentence-transformers==5.2.2

Only required libraries were imported and initialized.
A requirements.txt file is included for reproducibility.

Coding Environment

The solution was executed in a Jupyter Notebook environment.

To reproduce results:
1. Install dependencies from requirements.txt
2. Run the notebook from top to bottom
3. Ensure an active internet connection (required for sentence-transformers downloads)

Data Used

Only datasets provided on the competition page were used.

Target variable:
- adopted_within_07_days
- adopted_within_90_days
- adopted_within_120_days

Data Processing

All preprocessing steps are fully reproducible within the notebook.

Processing includes:
- Dropping duplicates using feature-engine
- Encoding categorical variables
- Aligning train and test feature columns

No postprocessing was performed outside the notebook.

Cross-Validation & Training Strategy

A 10-fold StratifiedKFold cross-validation strategy was used.

The fit_predict() function:
- Performs 10-fold stratified CV
- Trains estimator per fold
- Computes Log Loss and ROC-AUC
- Averages predictions across folds

Final prediction = mean of fold predictions

Accessibility

All libraries used are freely available.
No proprietary tools or paid services were used.

Exploratory Data Analysis (EDA)

An additional EDA notebook is included in the submission folder.
It contains:
- Target distribution insights
- Feature distributions
- Correlation analysis
- Modeling insights

Reproducibility Instructions

1. Install dependencies: pip install -r requirements.txt

2. Ensure internet connection is active
3. Run notebook from top to bottom

Expected runtime: ~1 hour 30 minutes.

Conclusion

This solution demonstrates a reproducible, modular, and production-ready pipeline for predicting early farmer adoption behavior.

Final Leaderboard Position: 9th Place